

MICAH BALDONADO

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Artificial Intelligence Engineering - Biomedical Engineering – GPA: 4.00/4.00

December 2025

Biomedical Engineering Master's Student Ambassador

Current Relevant Coursework: Machine Learning and AI for Engineers, Systems and Tool Chains for AI Engineers, Introduction to Deep Learning, Clinical Translations of AI

University of North Carolina at Chapel Hill & NC State University

Chapel Hill, NC

Bachelor of Science in Biomedical Engineering - GPA: 3.65/4.00

May 2024

EXPERIENCE

Ford Motor Company

Dearborn, MI (Hybrid)

Incoming AI/ML Intern

May 2025 – August 2025

- Selected for a 10-week summer internship on the AI/ML team within Ford's Global Data Insights and Analytics branch

PROJECTS

AI Emotional Companion – Empathetic Chatbot for Mental Wellness

Pittsburgh, PA

Team Lead & AI Engineer | React · FastAPI · OpenAI API · MongoDB

April 2025

- Led a 4-member team to design, build, and locally deploy a full-stack AI chatbot offering real-time, judgment-free emotional support; pitched the idea and managed team responsibilities
- Solely designed the AI agent architecture (emotion classification, psychoanalysis, user modeling, and RAG), integrating backend with frontend to deliver a functional demo

RESEARCH

Choroidal Analysis and Research (CAR) Lab

Pittsburgh, PA

Research Assistant

February 2025 - Present

- Working in a four-person team to develop interpretable deep learning methods for retinal image analysis; targeting submission to NeurIPS (A* conference)
- Applied a Canny edge detector and custom segmentation methods to estimate foveal thickness, narrowing down potential methods for fovea localization
- Developed vanilla saliency maps for interpretable fovea localization from B-scan OCT images to determine the best modality of saliency heatmaps for our team

Computational Biophotonics Lab

Chapel Hill, NC

Automated Behavior Analysis

June 2023 – May 2024

- Integrated machine learning tool Keypoint MoSeq into lab analysis pipeline, optimizing behavior labeling from manual to automated and enabling spatiotemporal comparisons across experimental conditions
- Developed MATLAB Guided User Interface, allowing users to create analysis boundaries within videos, enabling lab members to create a customized analysis of mice behavior depending on the experiment type
- Acknowledged in the preprint *Social threat alters the behavioral structure of social motivation and reshapes functional brain connectivity* (bioRxiv) for assistance with initial Keypoint MoSeq analysis

SKILLS

Programming Languages: Python, MATLAB, SQL, HTML/CSS, JavaScript, C++ (Arduino), Java

Machine Learning: Neural Networks, Ensemble Methods (XGBoost, Bootstrap Aggregation), Decision Trees, Random Forest, SVM, Clustering Techniques (K-Means, DBSCAN, Hierarchical), Dimensionality Reduction (PCA, t-SNE)

Software: PyTorch, Scikit-learn, Apache Spark, Google Cloud Dataproc, PostgreSQL, Neo4j, React, Matplotlib, Seaborn

Languages: English (Native), French (Intermediate), Filipino (Intermediate)